

BACLES, Romania

WMO: 154120

Lat: 44.4761N

Lon: 23.1131E

Elev: 315

StdP: 97.60

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-8.7	-13.8	1.2	-10.2	-11.6	1.4	-7.3	12.5	2.7	12.2	2.1	2.0	270	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	32.3	20.3	30.8	19.9	29.3	19.6	22.3	28.9	21.3	27.9	20.5	27.1	2.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	15.4	25.4	19.2	14.5	24.6	18.3	13.7	23.8	67.1	28.8	63.4	28.0	60.4	27.1	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	9.1	7.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-1.0	2.2	6.2	11.2	16.4	20.8	22.3	22.5	17.3	11.4	5.5	1.3		
	DBStd	9.09	4.54	4.72	4.88	4.69	3.25	2.97	2.87	3.18	3.95	4.28	4.36	4.35		
	HDD10.0	1188	342	221	134	42	1	0	0	0	0	35	146	269		
	HDD18.3	2930	600	452	375	216	76	10	3	6	66	214	385	527		
	CDD10.0	1702	0	2	17	78	199	324	381	389	221	80	11	1		
	CDD18.3	403	0	0	0	3	16	85	126	136	36	1	0	0		
	CDH23.3	3441	0	0	0	24	117	691	1127	1193	277	12	0	0		
	CDH26.7	1127	0	0	0	2	13	199	391	452	69	0	0	0		
(14) Wind	WSAvg	2.6	2.5	2.8	2.8	3.1	2.6	2.5	2.4	2.3	2.6	2.5	2.1	2.6		
(15) Precipitation	PrecAvg	567	31	29	35	52	61	58	61	36	57	41	40	50		
	PrecMax	1019	80	70	87	124	138	139	131	107	177	91	106	149		
	PrecMin	314	4	1	3	9	16	11	9	2	3	2	0	1		
	PrecStd	151	20	21	23	24	32	30	34	25	42	28	27	41		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	16.4	21.9	25.9	28.4	32.9	33.9	34.7	31.1	25.2	18.8	13.7		
		MCWB	8.0	10.0	12.4	15.8	19.6	20.5	21.9	20.2	18.5	15.6	12.5	8.8		
	2%	DB	8.8	13.0	18.5	23.4	26.2	30.8	32.1	32.5	28.6	22.7	15.4	10.7		
		MCWB	5.7	8.3	11.1	15.1	17.8	20.3	20.9	19.8	17.5	14.5	10.8	7.0		
	5%	DB	6.7	10.5	15.5	20.9	24.7	28.9	30.5	30.8	26.3	20.2	13.0	8.6		
		MCWB	4.2	7.0	9.6	13.2	17.0	20.2	20.3	19.5	17.0	13.5	10.3	6.1		
	10%	DB	4.9	8.3	13.2	18.6	23.0	27.3	28.9	29.2	24.2	18.0	11.2	6.6		
		MCWB	3.2	5.4	8.1	12.1	16.2	19.4	19.9	19.2	16.5	12.4	9.2	4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	10.8	14.0	16.7	20.5	22.6	24.6	23.8	20.5	16.2	13.1	10.0		
		MCDB	11.7	15.5	20.3	24.7	26.9	29.5	30.9	31.1	26.7	21.8	16.9	11.1		
	2%	WB	6.0	8.9	11.8	15.4	18.7	21.7	22.6	22.1	19.0	15.0	11.8	7.7		
		MCDB	8.5	12.5	17.0	23.0	24.5	28.3	28.8	28.9	25.2	20.7	14.4	10.4		
	5%	WB	4.4	7.2	10.1	14.0	17.7	20.9	21.3	21.0	17.8	14.0	10.6	6.2		
		MCDB	6.0	10.0	15.0	20.1	23.2	27.5	27.8	27.8	24.1	19.6	12.6	8.7		
	10%	WB	3.2	5.9	8.7	12.6	16.6	19.9	20.5	20.0	17.0	12.9	9.5	4.7		
		MCDB	4.7	7.9	12.6	17.6	21.8	26.2	27.1	26.4	23.2	17.1	11.3	6.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	7.1	8.5	9.7	9.8	10.7	11.2	11.1	10.0	8.6	6.1	5.9		
		MCDBR	8.6	10.3	12.5	12.3	11.9	12.4	12.8	12.9	12.8	11.9	9.3	8.3		
	5% WB	MCWBR	5.9	6.7	6.9	6.4	5.4	5.1	4.8	4.7	4.8	5.6	5.7	5.7		
		MCWBR	7.2	9.8	11.9	11.9	11.2	11.8	12.1	11.6	11.3	11.2	7.7	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.329	0.358	0.392	0.449	0.430	0.426	0.440	0.457	0.423	0.385	0.358	0.331		
		taud	2.315	2.239	2.217	2.070	2.191	2.239	2.206	2.152	2.216	2.315	2.337	2.339		
	Ebn at Noon		772	813	834	812	843	846	829	797	793	774	730	729		
		Edh at Noon	82	106	124	155	142	136	139	142	123	97	81	73		
(44) All-Sky Solar Radiation	RadAvg		1.39	2.20	3.44	4.68	5.75	6.43	6.41	5.66	4.04	2.60	1.46	1.15		
	RadStd		0.21	0.33	0.43	0.49	0.45	0.51	0.41	0.48	0.35	0.36	0.21	0.17		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (62 neighbors)		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	-159	+102	N/A	

Nomenclature: See separate page